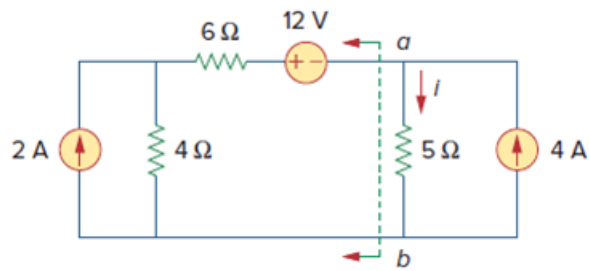


Problem

Obtain the Norton equivalent of the circuit in Fig. 4.116 to the left of terminals $a-b$. Use the result to find current i .

Figure. 4.116:



Step-by-step solution

Step 1 of 6

Circuit to find Norton equivalent resistance

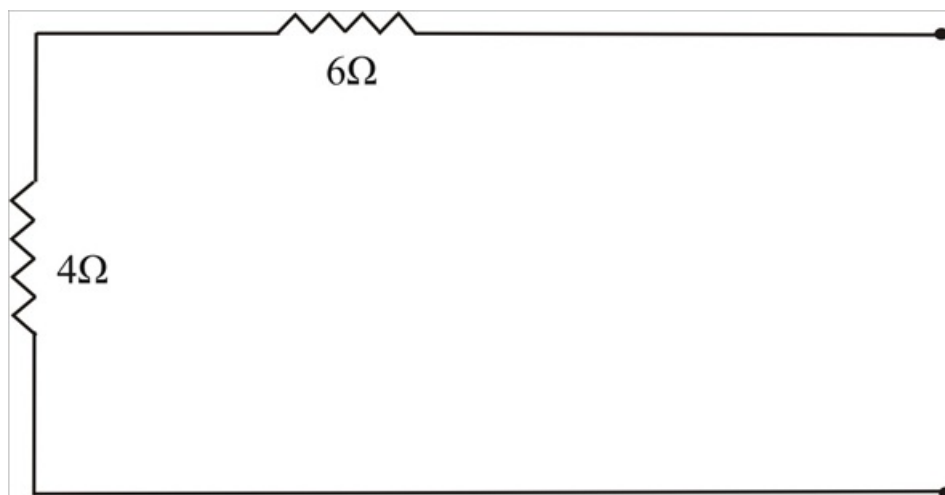


Fig-(a)

Step 2 of 6

From Fig-(a)

$$R_N = 6\Omega + 4\Omega$$

$$R_N = 10\Omega$$

Step 3 of 6

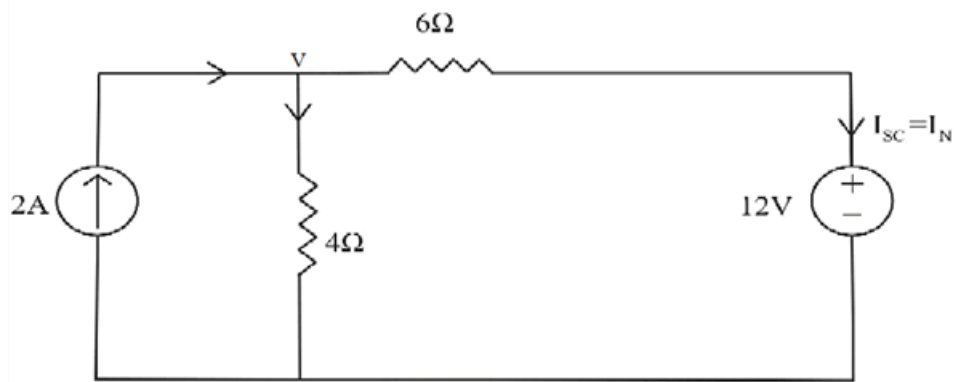


Fig-(b)

Step 4 of 6

From Fig-(b)

Writing nodal equation at top node;

$$2 + \frac{(12-V)}{6} = \frac{V}{4}$$

$$V = 9.6 \text{ V}$$

Therefore,

$$-I_N = \frac{(12-V)}{6}$$

$$-I_N = \frac{12-9.6}{6}$$

$$-I_N = 0.4$$

$$I_N = -0.4 \text{ A}$$

Step 5 of 6

Combining the Norton equivalent with the right hand side of the original circuit produces the circuit in Fig-(c).

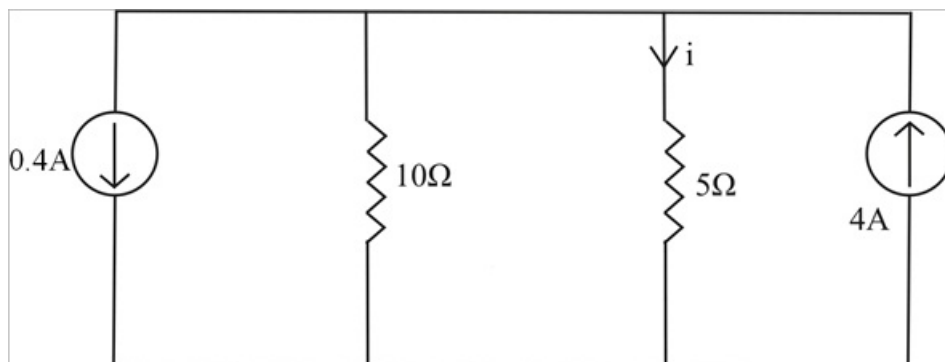


Fig-(c)

Step 6 of 6

$$i = \left[\frac{10}{(10+5)} \right] \times (4 - 0.4)$$

$$i = 2.4 \text{ A}$$

[Comments \(1\)](#)



Anonymous

As i goes down the 5 Ohm resistor, isn't the current division supposed to be $(5/(10+5))$?
